

Plan: HyMeKo-Gömb under the unrestricted (transductive) protocol on Epinions

2026-05-14

1 Scope and goal

Bracket the strict-protocol Gömb-Epinions number (0.9526 ± 0.0018) from above by measuring Gömb under the canonical transductive convention (test-edge signs participate in cycle σ -products). The strict number stays the headline; the new number sits in the supplement and quantifies the leakage path's contribution on Epinions specifically.

2 Hypothesis

Predicted lift over strict: $+0.01$ to $+0.05$ AUC. Lower bound from “Epinions structural prior already saturated” (small lift); upper bound from the Bitcoin-Alpha calibration (HSiKAN-Optuna jumps $\sim +0.05$ from strict-territory to transductive 0.9970).

3 Affected files

- `signedkan_wip/src/run_gomb_smoke.py` — add `-unrestricted-cycles` flag, swap `(e_tr, s_tr) → (g.edges, g.signs)` in the cycle-enumeration block when the flag is on. Cycle pool is constructed BEFORE the train/val split, so the test-edge signs leak into the σ -product features. Training labels and val/test metric computation are unchanged.
- `signedkan_wip/tests/test_gomb_unrestricted_flag.py` — new unit test. Verifies (i) the flag is plumbed, (ii) the cycle pool with the flag on enumerates strictly more cycles than with it off when there are test edges in the graph, (iii) the audit shuffle still flips the unrestricted run to a measurably different AUC (sanity).
- `signedkan_wip/experiments/run_gomb_epinions_unrestricted_2026_05_14.sh` — 5-seed runner using the same `v5_combined` config as the strict Epinions SOTA-break run.

4 CORE.YAML items touched

None. `signedkan_wip` is not listed in CORE.YAML; only the core crates (`hymeko_core`, `hymeko_query`, `hymeko_client`, `hymeko_daemon`, `parser`) and the spec / RTL files are protected.

5 Interface changes

Public: a single new CLI flag `-unrestricted-cycles` (default `False`). The existing strict protocol remains the default. No behaviour change for callers that do not set the flag. No internal API surface changes.

6 Test strategy

- *Unit* (1 file, ~ 3 tests): cycle-pool-size monotonicity, flag presence in the argparse, JSON output records the flag value.
- *Integration smoke*: 1-seed run on Bitcoin Alpha (~ 2 min) with the flag on. Verifies the pipeline doesn't crash, the cycle count grows vs strict, the AUC is non-degenerate.
- *Performance / measurement*: 5-seed Epinions run, same `v5_combined` config as the strict SOTA-break run, wall-time recorded per seed. Same RSS budget (~ 6 GB).

7 Performance budget

- Peak RSS: ≤ 8 GB (well under the 16 GB cap).
- Total wall time: ≤ 60 min for 5 seeds (strict baseline was ~ 33 min; unrestricted pool will be modestly larger because test-edge signs enter cycle enumeration, but cycle topology is identical, so wall time should rise $\leq 1.5\times$).
- GPU memory: ~ 5.5 – 6 GB.

8 Rollback path

The flag defaults to `False`; any call site that doesn't pass `-unrestricted-cycles` continues to run the strict protocol bit-for-bit. The unit test pins this default. To revert, delete the flag-handling block and the unit test — ≤ 30 LOC diff total.

9 Risk anticipation

- *Performance-contract preservation*. The cycle-pool enumeration already enforces `max_cycles`, top- k caps, and ABB pruning. Swapping the edge set from `(e_tr, s_tr)` to `(g.edges, g.signs)` preserves these caps because they are properties of the enumeration call, not of the edge set. Verified by reading the enumeration callsites at `run_gomb_smoke.py:464-515`.
- *Worst-case input*. Epinions: 131 828 vertices, 841 372 edges. Test-edge inclusion raises edge count by $\sim 25\%$ ($5/4$ of training). Cycle pool grows roughly proportionally; ABB cap and `topk=64` per-vertex keep pool size bounded.
- *Audit sanity*. The label-shuffle audit MUST still flip the unrestricted run to a different AUC (expected: AUC drops, but NOT to chance — the leakage path is now open, so shuffled signs still corrupt the prediction, just less dramatically than under strict). This is part of the test plan.
- *Empty-plan-dir hygiene*. If the experiment is abandoned, the plan dir is deleted before commit.

10 Decision tree

- Lift $\leq +0.01$: strict number is the result; the unrestricted number is supplementary noise. “Architecture saturates the structural prior; leakage path adds nothing.”
- Lift $+0.01$ to $+0.03$: moderate leakage contribution; goes in the methodology paper as an Epinions-specific quantification of the canonical convention’s information leak.
- Lift $\geq +0.04$: Gömb becomes the transductive leaderboard leader and the methodology paper has a stronger empirical contribution; the strict 0.9526 stays the headline because it is the structurally clean number.

11 Bottom line

A small, additive change to a non-core experiment script. Unit test pins behaviour; rollback is trivial. The result either strengthens the strict-protocol headline (by showing the leakage path adds little) or strengthens the methodology contribution (by quantifying the leakage path’s lift on Epinions).